Patterns, Pressure, and Forecasting

Team (replace with your names)

- Person A (Team Lead): Intro, hypothesis, conclusion & timeline
- Person B: Dataset overview, preprocessing plan, ethics & limitations
- Person C: Algorithms (FFT/QR/LU/clustering) & visualization plan

Deliverables (Part 1): 10-minute presentation (PDF) + contribution report (.txt)

Problem & Goal

Why study shelter occupancy and capacity?

- Shelter services report daily occupancy and capacity; persistent high pressure can signal resource constraints.
- Goal: quantify system-wide patterns, identify chronic high-pressure programs, and test for weekly/seasonal cycles.
- Practical value: support planning, staffing, and evidence-based resource allocation.

Hypothesis (H1)

Occupancy rate shows strong weekly seasonality and persistent high-pressure programs, and pressure differs by program/service type and by capacity type (bed vs room).

Data Sources

What we will use and why it fits the question

Primary dataset (program-level, daily):

Daily Shelter & Overnight Service Occupancy & Capacity (Toronto open data)

https://data.urbandatacentre.ca/fr/catalogue/city-toronto-daily-shelter-overnight-service-occupancy-capacity

Direct CSV export example (2024):

https://ckan0.cf.opendata.inter.prod-toronto.ca/datastore/dump/fc409fd7-0348-49d7-bba9-70ac1a8c727c

How the daily number is measured

City reporting is a daily snapshot around 4 a.m.; updates are posted daily except weekends/holidays.

https://www.toronto.ca/city-government/data-research-maps/research-reports/housing-and-homelessness-research-and-reports/shelter-census/

Optional for mapping later:

Shelter locations (GIS points):

https://data.urbandatacentre.ca/fr/catalogue/city-toronto-hostel-services-homeless-shelter-locations

Preprocessing Plan

Clean, transform, and build analysis-ready features

Steps

- Select time window: 2021-2024 (consistent schema)
- Parse dates; standardize program/service categories
- Handle missing values and capacity=0 safely
- Create derived metrics (rates and flags)
- Aggregate views: system-wide, by category, per program

Key derived metrics

- occupancy_rate_actual = occupied / actual_capacity
- occupancy_rate_funding = occupied / funding_capacity
- pressure flags: days >= 90%, 95%, 100%
- time features: day_of_week, month, weekend
- optional: lag features for forecasting

Ethics & Limitations

Bias, sensitivity, and responsible communication

Key considerations

- Sensitive topic: present results in aggregate; avoid stigmatizing shelters or neighborhoods.
- Confidentiality and omissions can affect representativeness (e.g., certain shelter types may be excluded in some reporting contexts).
- Weekend/holiday reporting cadence can distort naive day-of-week analyses; we will treat update gaps carefully.
- Capacity definitions matter (bed vs room; funding vs actual). Misuse can over/understate pressure.
- We will document assumptions, preprocessing decisions, and uncertainty.

Algorithms

Methods to test the hypothesis and summarize patterns

1) Seasonality detection (FFT)

Apply FFT to system-wide occupancy_rate_actual to detect weekly periodicity and compare strength across years.

- Regression: predict next-day occupancy rate using day-of-week, month, and lag features.
- Clustering: group programs into pressure profiles (chronically full, seasonally spiky, lower utilization).

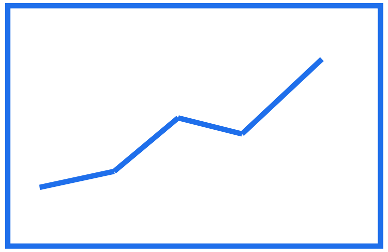
Nice-to-use math (for justification):

QR factorization for stable least-squares fitting; LU decomposition for efficient linear system solves.

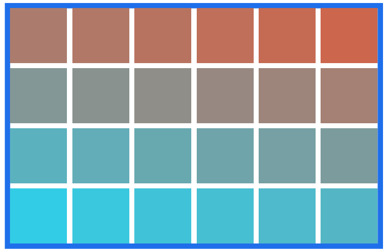
Planned Visualizations

How we will communicate patterns and findings

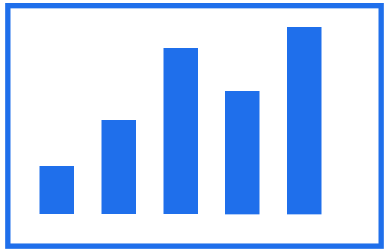
Core plots



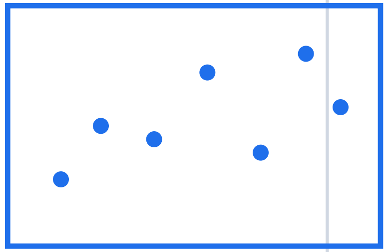
System occupancy
over time



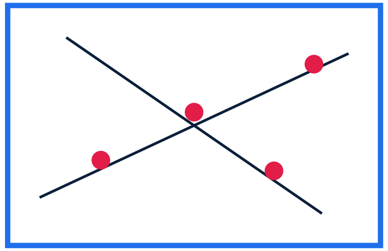
Day-of-week x
month heatmap



% days >=95%
by program



Cluster plot
(profiles)



Optional map
(pressure category)

We will prioritize clear, non-stigmatizing framing: focus on system-level pressure and program profiles, not ranking neighborhoods.

Plan, Roles & Next Steps

What we will do after Part 1

Work plan (example 2-3 week timeline)

- Week 1: download + validate data; preprocessing pipeline; derived metrics (rates/flags).
- Week 2: exploratory analysis + baseline visuals; FFT seasonality; initial regression.
- Week 3: clustering program profiles; refine visuals; interpret results + limitations.

Role split (edit names)

- Person A: narrative, hypothesis, final integration & presentation flow
- Person B: data sourcing, preprocessing, ethics/limitations write-up
- Person C: algorithms (FFT/QR/LU), clustering, visualization draft set

Deliverables after Part 1

- Clean dataset + feature table
- Model outputs + evaluation notes
- Final visuals + interpretation
- Final report / presentation

References

Datasets and documentation (accessed Feb 3, 2026)

Daily Shelter & Overnight Service Occupancy & Capacity (catalogue):

https://data.urbandatacentre.ca/fr/catalogue/city-toronto-daily-shelter-overnight-service-occupancy-capacity

Direct CSV export example (2024):

https://ckan0.cf.opendata.inter.prod-toronto.ca/datastore/dump/fc409fd7-0348-49d7-bba9-70ac1a8c727c

City documentation (Shelter census / daily usage):

https://www.toronto.ca/city-government/data-research-maps/research-reports/housing-and-homelessness-research-and-reports/shelter-census/

Optional: Shelter locations (GIS):

https://data.urbandatacentre.ca/fr/catalogue/city-toronto-hostel-services-homeless-shelter-locations